

Multiple Choice (10 marks)

1. Find the inverse Laplace transform $L^{-1}\{1/(s-1)^2\}$, using the convolution.
 - (a) $t^2 e^t$
 - (b) $t e^{2t}$
 - (c) e^{-t}
 - (d) $t^3 e^t$
 - (e) te^t
2. Evaluate ∇f at the point (2, 3, 5). The scalar field f is given by $f(x_1, x_2, x_3) = 2 \sin x_1 - x_1^2 x_2 x_3 + x_1 e^{x_2}$
 - (a) (40.74, -20.17, 12)
 - (b) (-30.74, 10.17, -2)
 - (c) (0, 40.34, 24)
 - (d) (30.74, 10.17, 2)
 - (e) (40.74, 20.17, -12)
3. The current through an inductor with inductance $L = 10^{-3} \text{ henry}$ is given as $i_L(t) = 0.1 \sin 10^6 t$. Find the voltage $v_L(t)$ across this inductor.
 - (a) $1000 \sin 10^6 t$
 - (b) $100 \sin 10^6 t$
 - (c) $50 \cos 10^6 t$
 - (d) $1000 \cos 10^6 t$
 - (e) $0.1 \cos 10^6 t$
4. Find the inverse Laplace transform $f(t) = L^{-1}\{\log[(s+1)/(s-1)]\}, s > 1$.
 5. $(1 - e^{-2t}) / t$
 6. $(2 \sinh t) / t^2$
 7. $(2 \tanh t) / t$
 8. $(2 \sinh t) / t$
 9. $(2 \tanh t) * t$

In a lap winding dc machine number of conductors are 100 and number of parallel paths are 10. Find the average pitch

- (a) 150
- (b) 100
- (c) 200
- (d) 5
- (e) 10

True / False (10 marks)

Answer True or False

6. True or False: Gross errors are primarily caused by human mistakes during measurements or calculations.
7. True or False: A dc motor requires 10 kilowatts to supply 10 horsepower, resulting in an efficiency of 85%.
8. True or False: The convolution of $f(t) = t$ and $g(t) = t^2$ results in $f(t) * g(t) = t^4 / 12$.
9. True or False: The a-c voltage between rings of a 1,500-kw six-phase synchronous converter is 175 volts.
10. True or False: XOR and XNOR gates alone can function as universal gates without the need for additional gate types.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the required capacitive reactance to achieve unity power factor for a 220-volt, 60-hertz, 10-hp single-phase induction motor operating at 86% efficiency and 90% power factor. Discuss your reasoning and calculation steps.
12. Calculate the volume of a storage container needed for 1500 lbm of ethane (C_2H_6) at 160^a and 1100 psig, and discuss the factors influencing your calculations.

End of Paper